

UX Case Study: Redesigning the IRCTC Train Ticket Booking Experience

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Executive Summary

IRCTC (Indian Railway Catering and Tourism Corporation) handles over 1.5 million bookings daily. Despite its scale, the user experience is plagued by high cognitive load, technical friction, and a lack of emotional design. This research document identifies critical pain points in the booking flow, seat selection, and error handling, proposing a human-centric redesign grounded in established UX principles.

SECTION 1 — The Booking Flow (Search → Seat Selection → Payment)

Current State Analysis

The current flow requires users to navigate a “login wall” before they can even browse availability. The search interface is cluttered with ads and technical jargon (e.g., “GN”, “TQ”, “PT”).

Step	User Action	Current Pain Point
1	Search	Mandatory login before search; confusing station codes.
2	Train Selection	Cluttered UI with intrusive ads camouflaging critical data.
3	Review	“Tatkal chaos” during peak hours; system crashes under load.
4	Payment	Drop-offs due to session timeouts and lack of “saved” states.

Conscious vs. Subconscious Pain Points

- **Conscious Pain Points:** The “Login Wall” creates immediate friction. Tatkal booking (10 AM/11 AM) is a high-stress “lottery” where users face repeated `Error 503` or session timeouts [1]. Cryptic error messages like “Booking not available for the next hour” offer no explanation or recovery [2].
- **Subconscious Pain Points:** High **Cognitive Load** due to visual noise. **Trust Erosion** occurs when the system fails at the final payment step. The lack of **Progressive Disclosure** overwhelms first-time users with technical abbreviations.

Root Cause Analysis

The system prioritizes backend logic and security protocols (anti-bot measures) over user-centric design. The legacy architecture struggles with peak-hour concurrency, leading to “Stress-Use” scenarios where the UI fails exactly when the user needs it most.

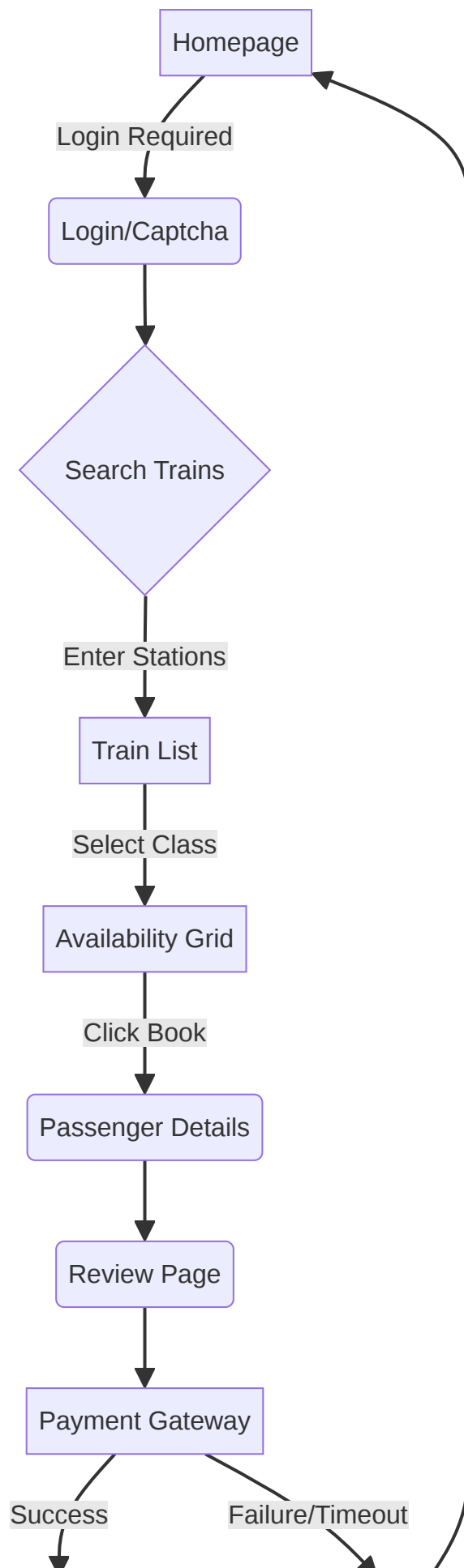
Proposed Fixes

- **Fix 1: Search-First Access.** Allow users to search and view availability without logging in. Only require login at the “Passenger Details” stage.
- **Fix 2: Natural Language Search.** Replace cryptic station codes with city-based suggestions (e.g., “Mumbai” shows all city stations).
- **Fix 3: Progressive Disclosure.** Hide advanced quotas (Tatkal, Ladies, Divyang) behind a “Filter” menu to reduce initial clutter.

Applied UX Principles

- **Error Prevention (Nielsen #5):** Real-time validation of station names to prevent “No trains found” errors.
- **Cognitive Load:** Reducing the number of choices on the homepage.
- **Progressive Disclosure:** Showing only necessary information at each step.

Current Booking Journey Flow



SECTION 2 — Seat Selection UI

Current State: The Grid Problem

The current “Seat Selection” is not a spatial map but a textual grid of dates and availability. The “Coach Layout” view is often hidden or unreadable on mobile devices, making it impossible for users to choose seats based on physical location (e.g., near doors, away from toilets).

Pain Points & Root Cause

- **Spatial Confusion:** Users cannot visualize where they are sitting.
- **Information Hierarchy:** Price, availability, and class are given equal visual weight, leading to scanning fatigue.
- **Root Cause:** The UI is a direct reflection of the database structure (rows and columns) rather than a mental model of a train coach.

Proposed Redesign Rationale

Introduce an **Interactive Coach Map**. This leverages **Spatial Mapping** to help users understand their environment.

Feature	Current State	Proposed Redesign
Layout	Text-based date grid	Visual coach map with seat icons
Interaction	Select date, then check availability	Click seat to select; instant price update
Hierarchy	Cluttered, small fonts	High-contrast icons and clear “Book” CTA

Applied UX Principles

- **Affordance:** Buttons that look like seats to invite interaction.
- **Spatial Mapping:** Aligning the digital interface with the physical layout of a train.

- **Information Hierarchy:** Using size and color to highlight “Available” vs. “Booked” seats.
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SECTION 3 — Error States & Recovery

Current: The “Dead End” Experience

Errors in IRCTC are often technical and punitive. If a payment fails or a session times out, the user is kicked back to the homepage, losing all progress.

Pain Points

- **Cryptic Codes:** “Error 102: Transaction Failed” provides no context.
- **Lack of Recovery:** No “Retry” button for failed payments; users must re-enter passenger details.
- **Tatkal Timeouts:** During peak load, the app often freezes without feedback, leaving users in limbo [3].

Proposed Redesign

- **Plain-Language Errors:** “We couldn’t reach your bank. Your money is safe.”
- **Recovery CTAs:** A “Retry Payment” button that preserves the booking session for 5 minutes.
- **Session Preservation:** Auto-save passenger details in local storage to prevent data loss on crash.

Applied UX Principles

- **Error Recovery (Nielsen #9):** Helping users recognize, diagnose, and recover from errors.
 - **Graceful Degradation:** Providing a simplified “text-only” flow during peak Tatkal hours to ensure functionality over aesthetics.
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Overall Impact Summary

Metric	Current State	Target Post-Redesign
Booking Time	4-6 Minutes	2 Minutes
Error Rate	High (Peak Hours)	Low (via session preservation)
User Trust	Low (Frustration-heavy)	High (Reliable & Transparent)
Cognitive Load	High (Visual noise)	Low (Guided flow)

Citations & References

1. **Reddit Community Discussion:** “Tatkal booking keeps failing with an error.” [Source](#)
2. **Economic Times:** “IRCTC down? Tatkal booking users complain repeated errors.” [Source](#)
3. **UX Planet Audit:** “UX audit of IRCTC booking flow presented to the IRCTC team.” [Source](#)
4. **App Store Reviews:** “3rd class app and useless app, poor standard.” [Source](#)